

Experiment 4: Simple E-Commerce Application Using React Router

App.js

```
import React from "react";
import {
  BrowserRouter,
  Routes,
  Route,
  Link
} from "react-router-dom";

function Home() {
  return <h2>Home Page</h2>;
}
function Products() {
  return <h2>Products Page</h2>;
}
function Cart() {
  return <h2>Cart Page</h2>;
}
function Contact() {
  return <h2>Contact Page</h2>;
}
function Login() {
  return <h2>Login Page</h2>;
}
function NotFound() {
  return <h2>404 Page Not Found</h2>;
}

function App() {
  return (
    <BrowserRouter>
      <nav>
        <Link to="/">Home</Link> |
        <Link to="/products">Products</Link> |
        <Link to="/cart">Cart</Link> |
        <Link to="/contact">Contact</Link>
      </nav>

      <Routes>
        <Route path="/" element={<Home />} />
        <Route path="/products" element={<Products />} />
        <Route path="/cart" element={<Cart />} />
        <Route path="/contact" element={<Contact />} />
        <Route path="/login" element={<Login />} />
        <Route path="*" element={<NotFound />} />
      </Routes>
    </BrowserRouter>
  );
}

export default App;
```

Output

- Home Page, Products Page, Cart Page, Contact Page and Login Page are displayed according to navigation.
- Invalid URL displays: 404 Page Not Found.

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Aim

1. To understand routing in React applications using React Router.
2. To implement navigation between pages without reloading the browser.
3. To create Home, Products, Cart, Contact and Login pages.
4. To demonstrate route handling including an invalid URL / 404 page.

Procedure

1. Create a React application using Create React App.
2. Install react-router-dom.
3. Create Home, Products, Cart, Contact and Login components.
4. Import BrowserRouter, Routes, Route and Link.
5. Configure navigation links and routes.
6. Run the application using npm start and verify navigation.

Sample Input

- Click Home, Products, Cart, Contact and Login links.
- Open an invalid URL to test the 404 route.

Sample Output

- Home Page, Products Page, Cart Page, Contact Page and Login Page are displayed according to navigation.
- Invalid URL displays: 404 Page Not Found.

Result

Thus, a simple E-Commerce application was successfully developed using React Router and navigation between multiple pages was demonstrated.

Product ID : 101

Invalid URL

404 Page Not Found

Result

Thus, a simple E-Commerce application was successfully developed using React Router. The application demonstrated client-side routing, navigation between multiple pages without page refresh, implementation of protected routes, dynamic routing using URL parameters, and handling of invalid URLs through a 404 page.

